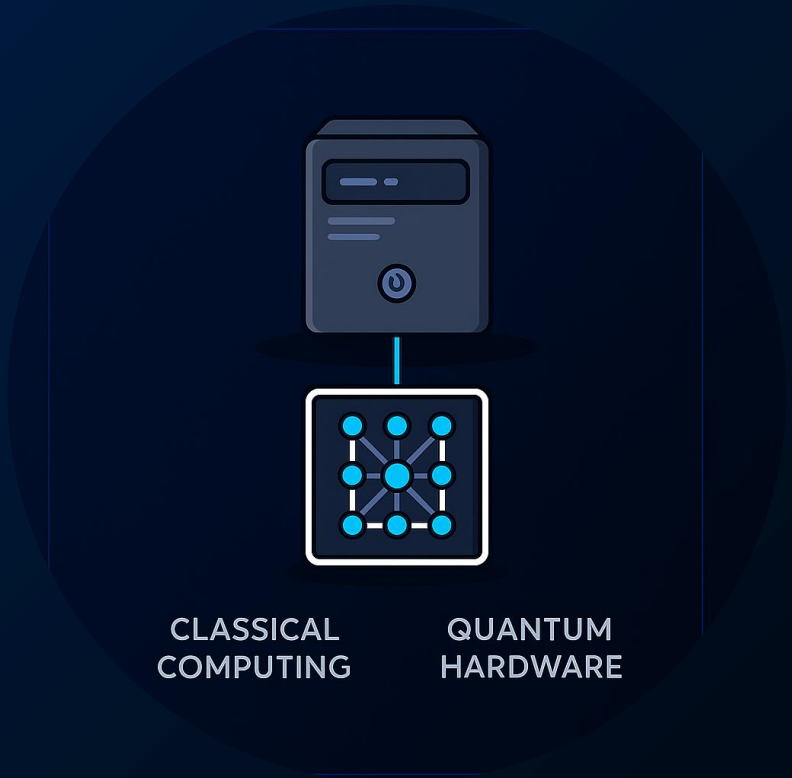
Boston Dynamics - AI and Robotics

Track 1: Object Pose Estimation

BlendED

Hybrid Quantum–Classical CNN for Image Segmentation

Team 2: Kangmin Ha, WangZhenqi, YeShengqing





Introduction

Methods

Results Analysis

Discussion

Introduction

- Abstract
- Background
- Research Methods

Introduction

BlendED

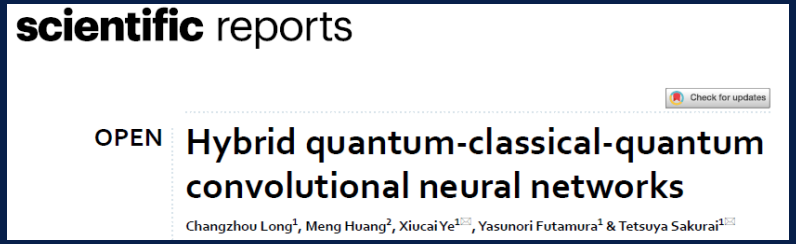
- Abstract

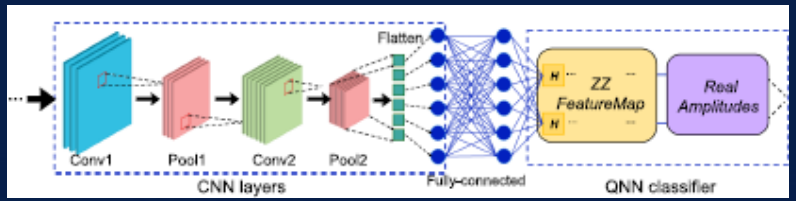
The basic pipeline extracts the point cloud for each object from the segmentation output as in the existing work, and then estimates the 6D pose by ICP-based matching. However, the problem is that the reliability of the segmentation stage determines the overall pipeline quality. Therefore, we propose a hybrid quantum-classical structure that reduces the false positive by trialing the mask around each patch using the candidate mask obtained with Classical Convolutional Neural Network and enhanced by Quantum Convolutional Neural Network. It is expected to be combined with classical hardware in the current generation of quantum hardware to improve segmentation classification quality.

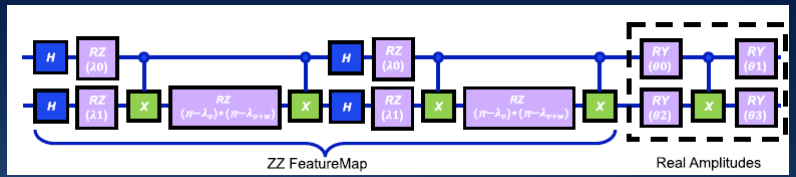
Introduction

- Backgrounds
  1. Classical CNN Method
     - Convolutional Neural Network
     - Controlling weights
  2. QCNN Layer
     - Quantum Circuit
     - Endcoding
     - Parametrized Quantum Circuit
  3. Application
     - QCNN layer for segmentation

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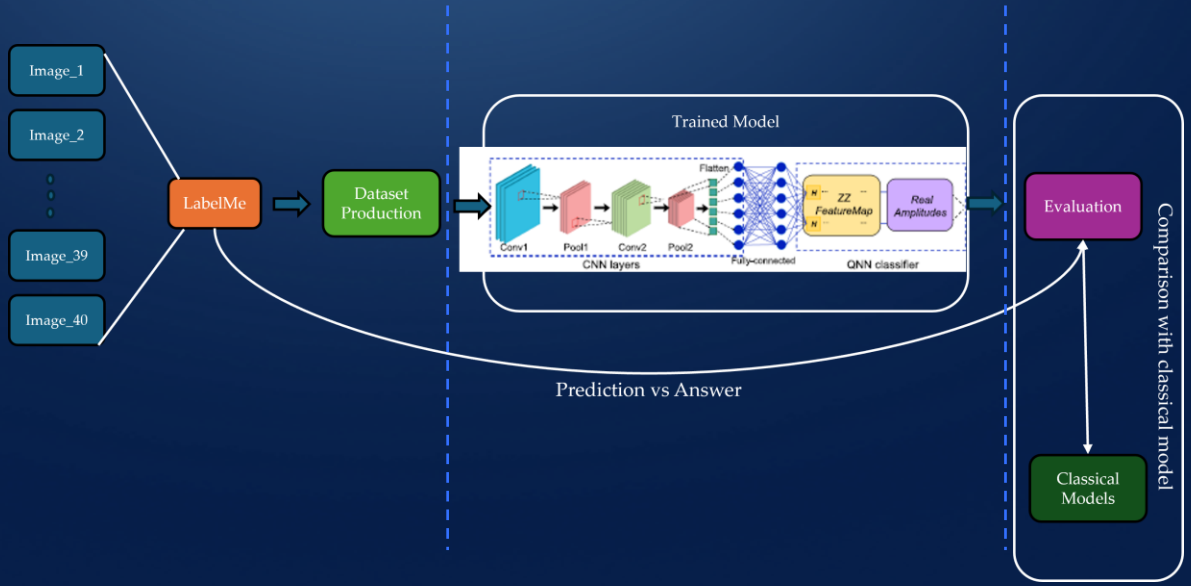




Long, C., Huang, M., Ye, X., Futamura, Y., & Sakurai, T. (2025). *Hybrid quantum-classical-quantum convolutional neural networks*. **Scientific Reports, 15**, 31780. https://doi.org/10.1038/s41598-025-13417-1

Introduction

- Research Methods



1. Classical Models

- YOLO model
- U - Net CNN model

1. QCNN Hybrid Model

- QCNN Layer on classification, but also on **segmentation**

Methods

- YOLO
- U -Net
- Hybrid QCNN

Methods

- Outline

Review of Existing Classical Models

Review of YOLO & U-Net CNN

Comparison

Milestone for CNN Extension

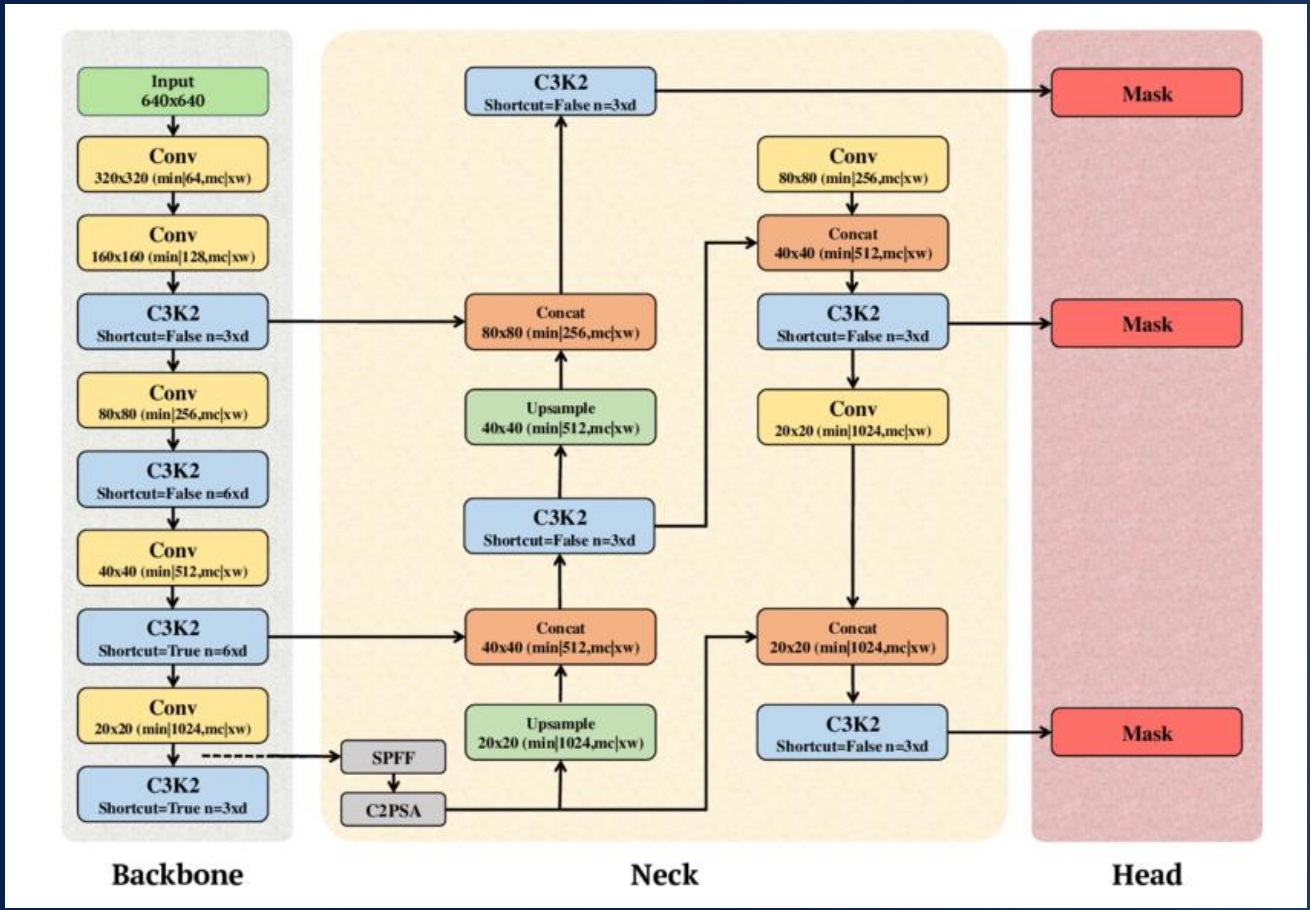
Hybrid QCNN -CNN Model

Hybrid model

Pure CNN model

YOLO

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YOLO-Classification

BlendED



Accuracy：0.989

YOLO-Segmentation

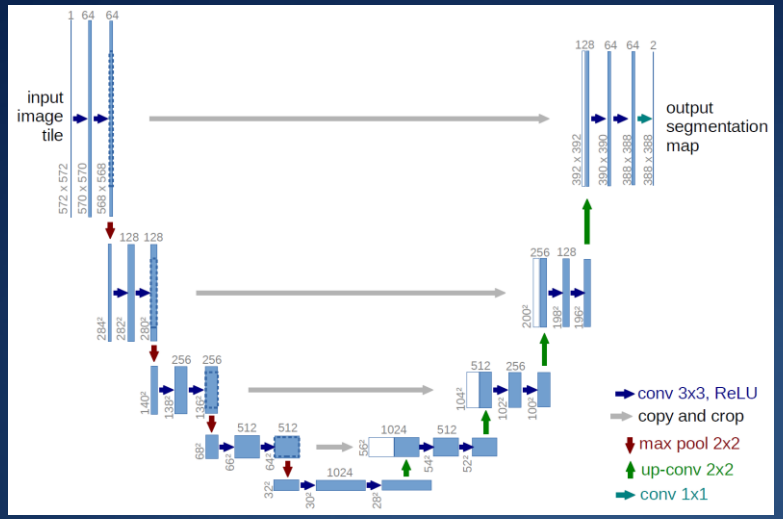
BlendED



Accuracy：0.799

U-Net (compare with YOLO)

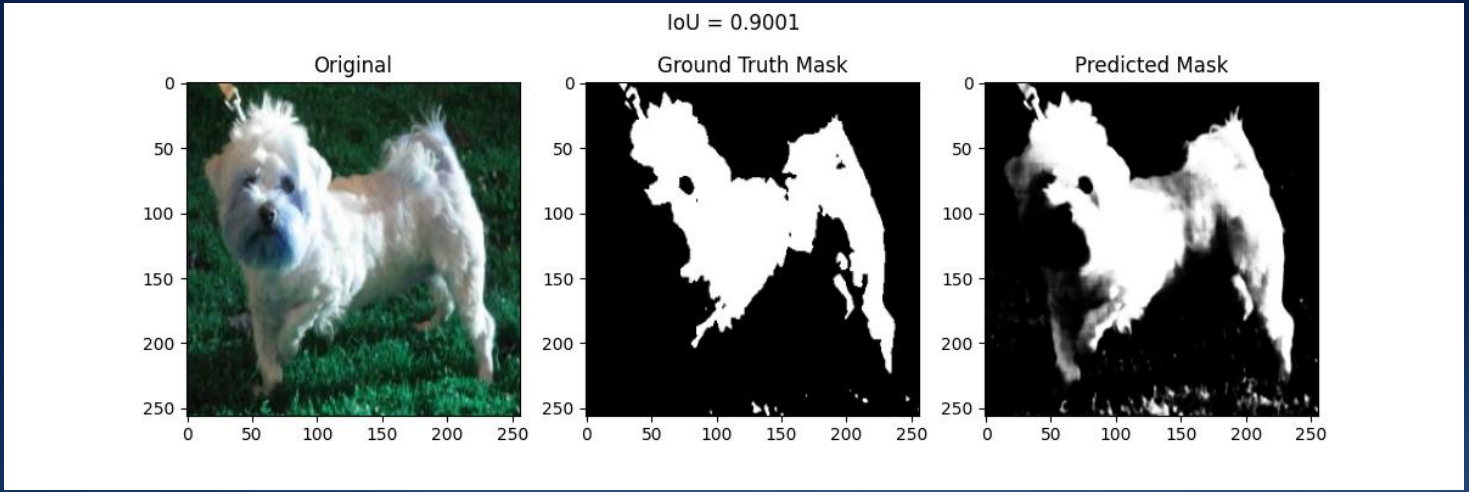
- Pixel-level output is more refined
- More details are retained
- More suitable for small sample tasks
- The structure is lightweight and easy to expand



U-Net Segmentation

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IoU = 0.9001



U-Net Classification

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```
Overall Accuracy: 0.8886
Class 'boxs' Accuracy: 0.6667
Class 'cats' Accuracy: 0.8843
Class 'dogs' Accuracy: 0.8943
```



BlendED

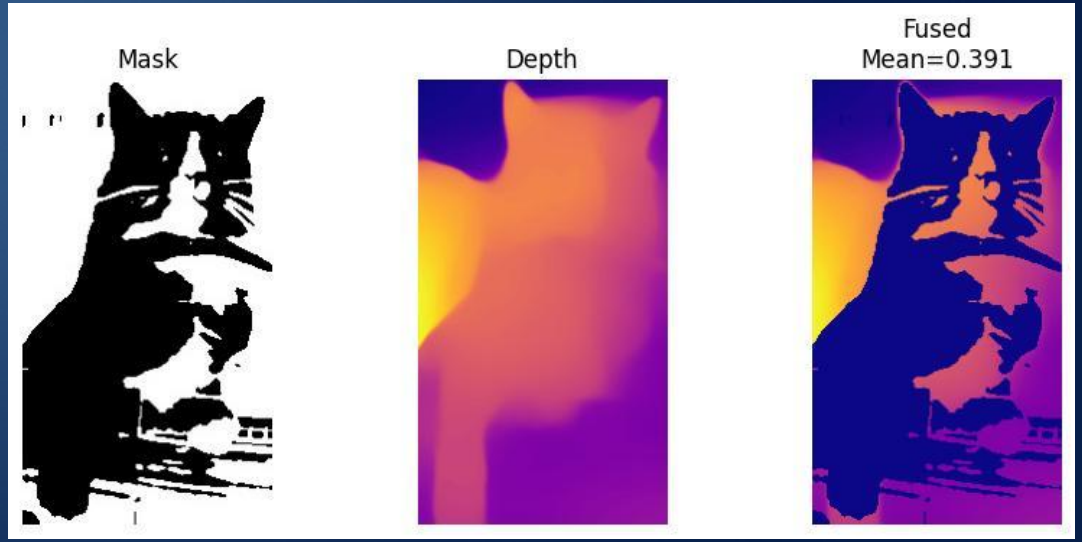




- The sample size is small.
- Confused with other categories
- The impact of the split mask

U-Net Depth estimation

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Close to 1: further

Close to 0: closer

Hybrid QCNN Model

BlendED

- Dataset

Image Files

LabelMe

Polygons

[40 total images]



Rasterize Mask

Segmentation ID:

Background = 0 (Black)

Small Book = 1 (Blue)

Mid Book = 2 (Green)

Big Book = 3 (Red)

Classification ID:

Small Book = 0

Mid Book = 1

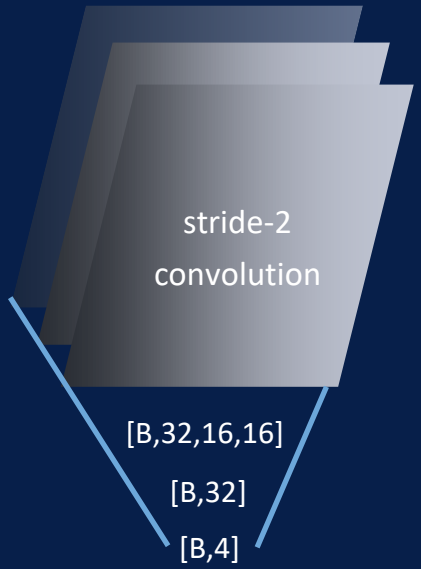
Big Book = 2

Hybrid QCNN Model

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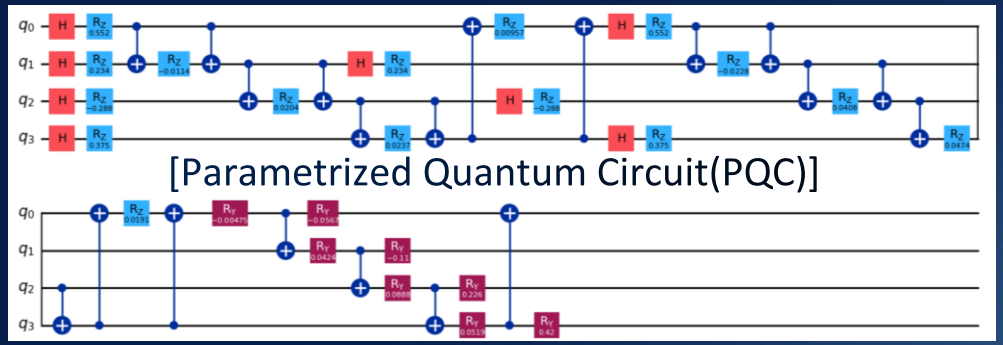
- Model

[CNN Backbone]



[QCNN]

[ZZ Feature Map]



PQC is also known as Ansatz

[Segmentation]

Broadcasting
[B,4,16,16]
torch.cat
[B,32,16,16]
ConvTranspose2d
[B,4,128,128]

[Classification]

Linear Layer
ReLU
Linear Layer
[B,3]
(dominant box
prediction)

Hybrid QCNN Model

BlendED

- Train
  - Based on pixel ID distribution, weights are applied to each segment IDs
  - Loss Function:
    - Usage of CrossEntropy
    - Total Loss = 0.4*ClassLoss + 0.6*SegmentationLoss
    - 40 Epoch
  - Forward <-> Backward:
    - Usage of quantum circuit
    - Getting gradient of parameters is contemporary named problem

Hybrid QCNN Model

BlendED

- Train
  - Spec:
    - 128 x 128 image
    - 4 Segmentations, 3 Classifications
    - 4 qubit circuit, 16 parameters
    - 40 epochs, at least two gradient calculations in each epoch

When it comes to uploading to IBM quantum platform,
it sends each QCs for 16 parameters x 2(gradient calculation) x 40 epoch= 1280 jobs
IBM platform cuts out long - large waiting queues, since it would occupy the platform several days.
(This kind of trouble often happens in QCNN field)

Hybrid QCNN Model

BlendED

- Train

  When it comes to uploading to IBM quantum platform,

  it sends each QCs for 16 parameters x 2(gradient calculation) x 40 epoch= 1280 jobs

  IBM platform cuts out long - large waiting queues, since it would take several days.

This is why prior works are mostly done for classification, not for segmentation.

However, to extract the quantum potential, we use:

- Qiskit - aer
  - Qiskit primitives, backend

Hybrid QCNN Model

BlendED

- Results

Evaluation:

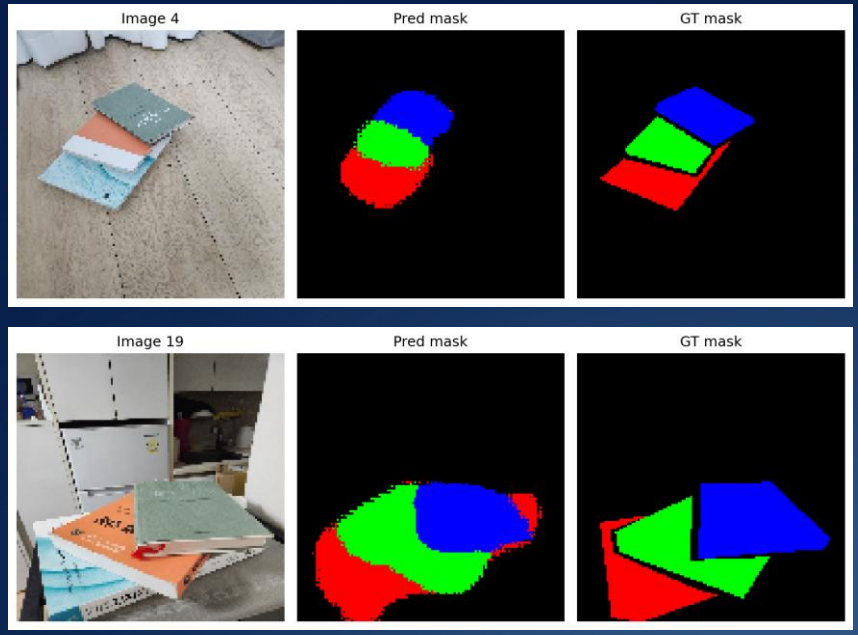
Sample Batch of 30 Images
->Prediction of pixel and class

📊 Classification accuracy: 90.00%

📊 Pixel accuracy: 94.36%

📊 Mean IoU: 70.56%

<Prediction Sample Images>

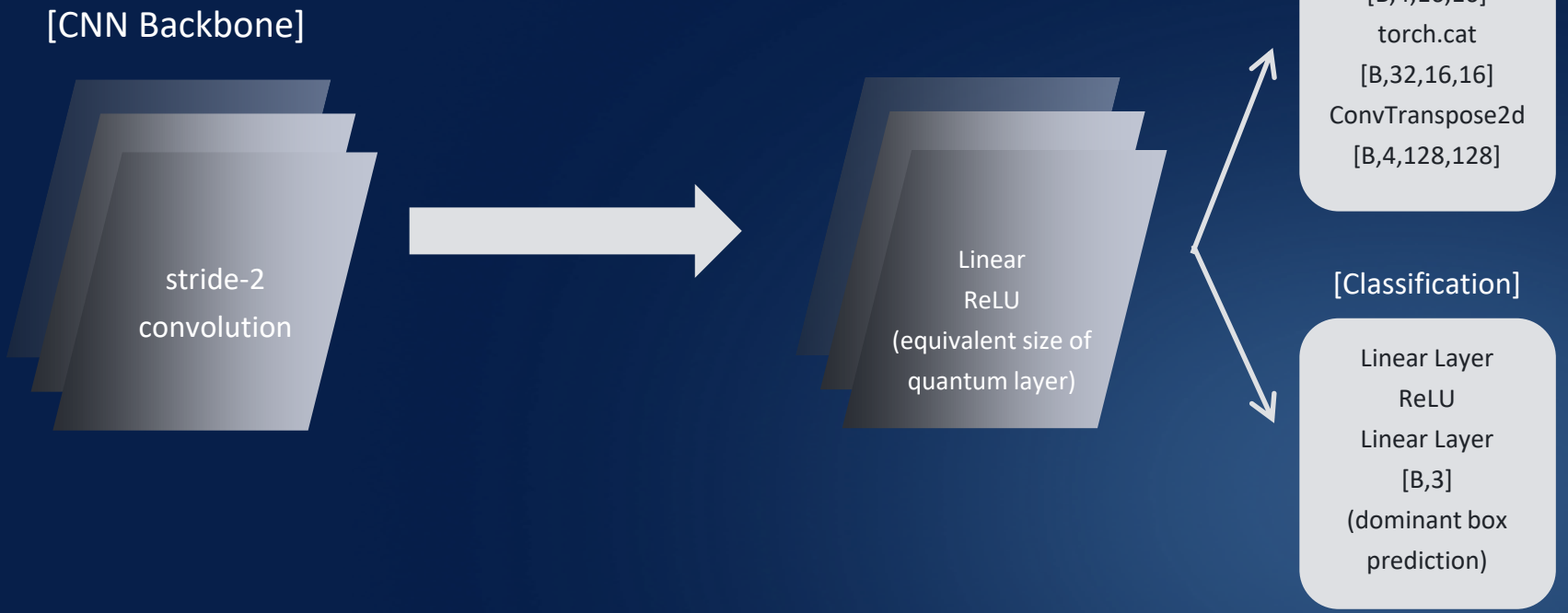


Results Analysis

- Comparison
- Reasoning
- Further Works

Comparison

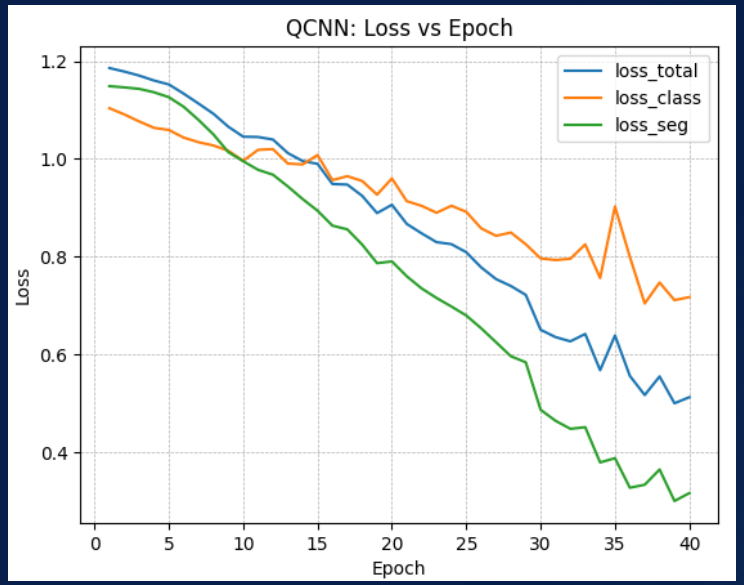
- Classical CNN-Only model

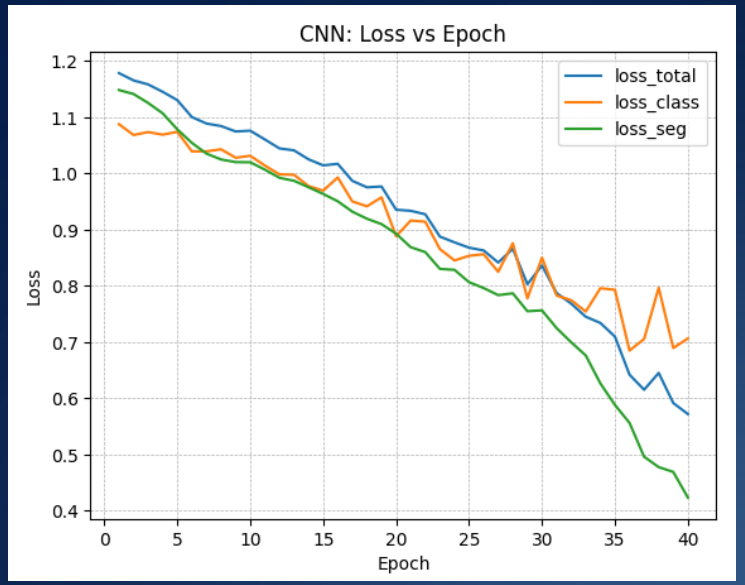


Comparison

BlendED

Hybrid QCNN model VS Classical CNN-Only model

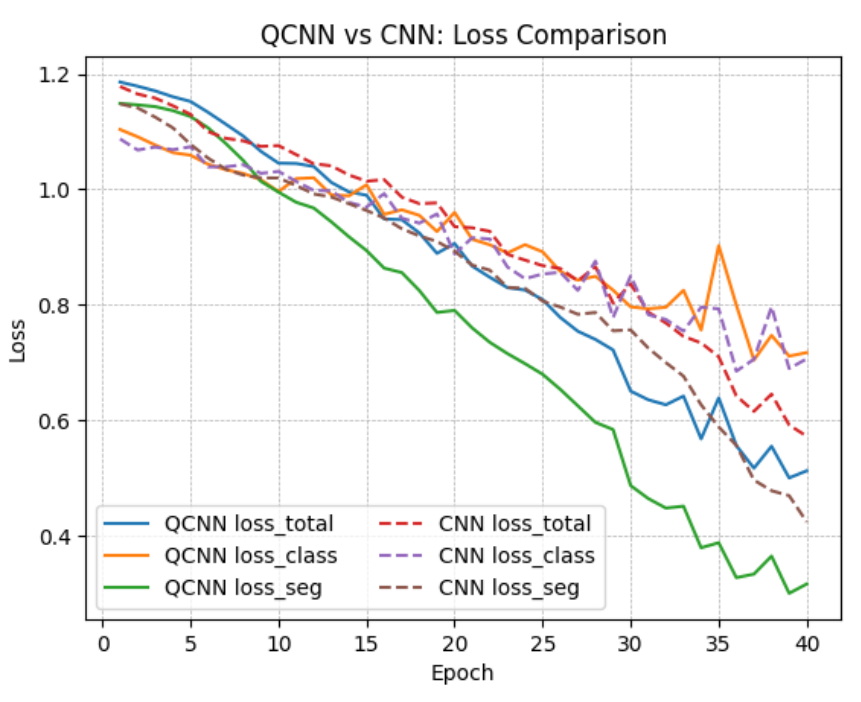




Comparison

BlendED

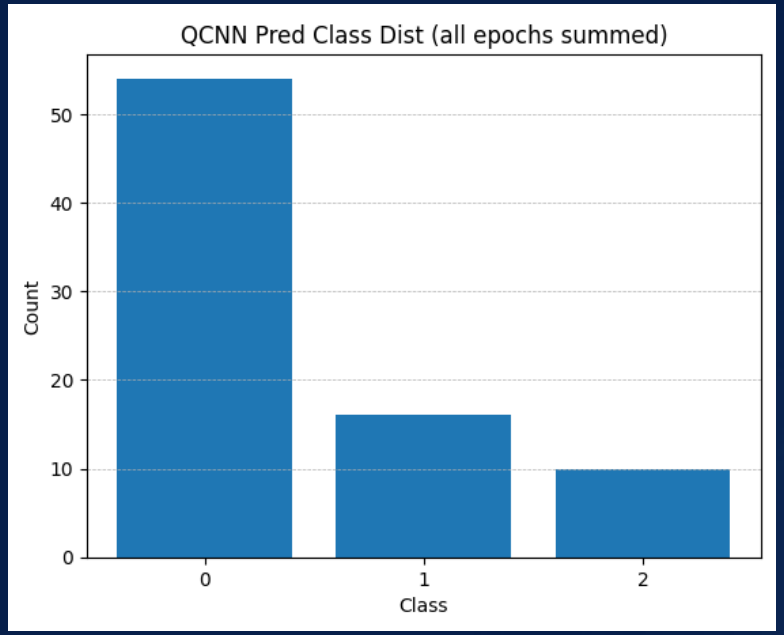
Hybrid QCNN model VS Classical CNN-Only model

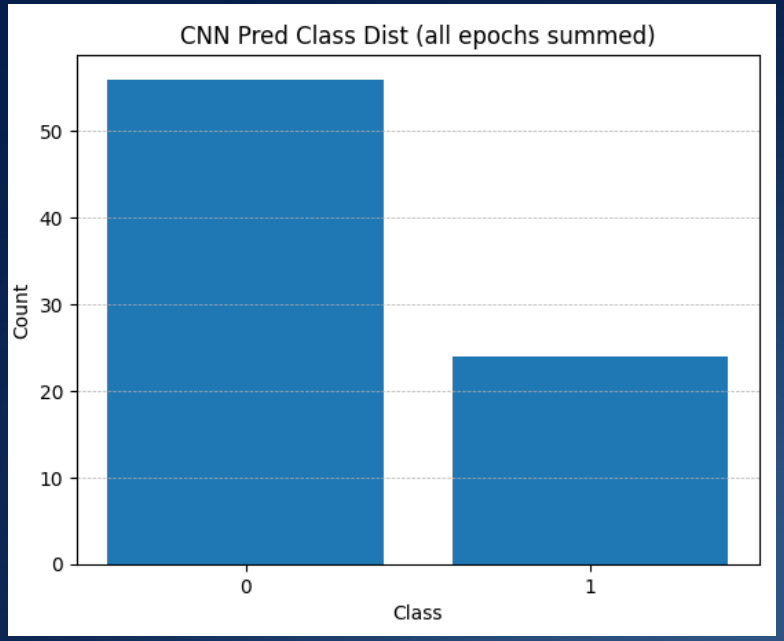


Comparison

BlendED

Hybrid QCNN model VS Classical CNN-Only model





Comparison

BlendED

Hybrid QCNN model VS Classical CNN-Only model

<Exact Ratio>

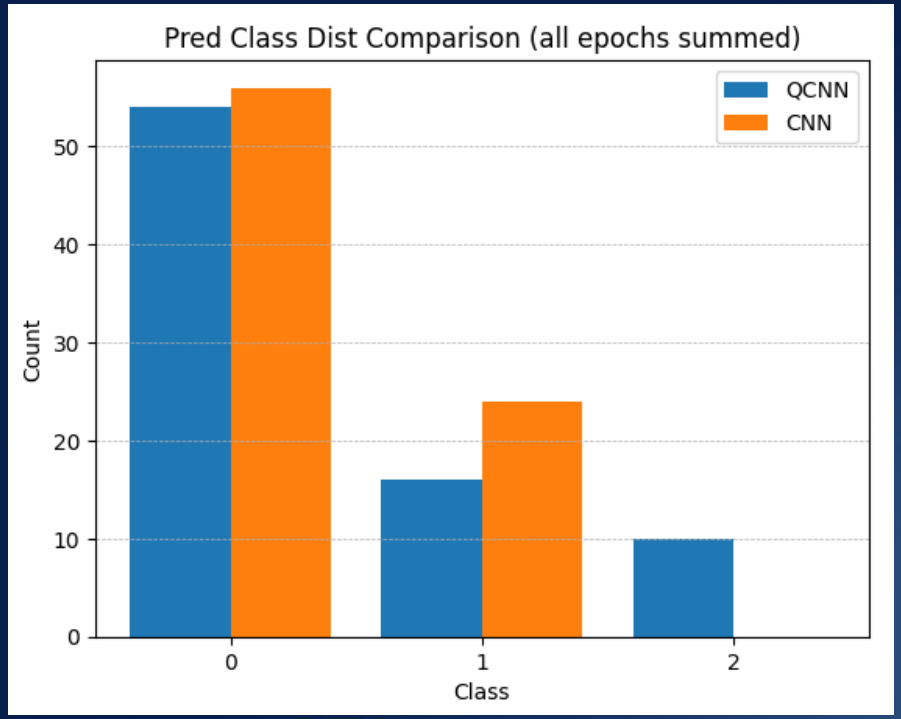
ID : number of classes

0(small): 4.33

1(mid): 3

2(big): 2.66

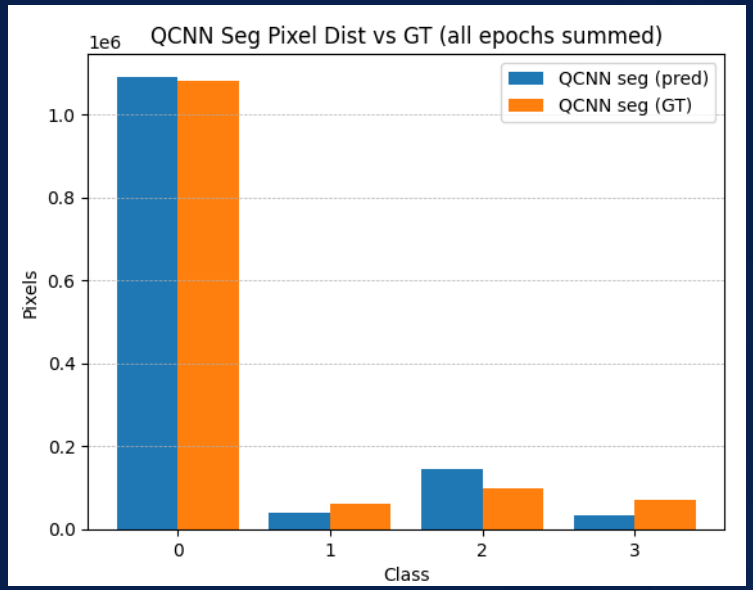
/10

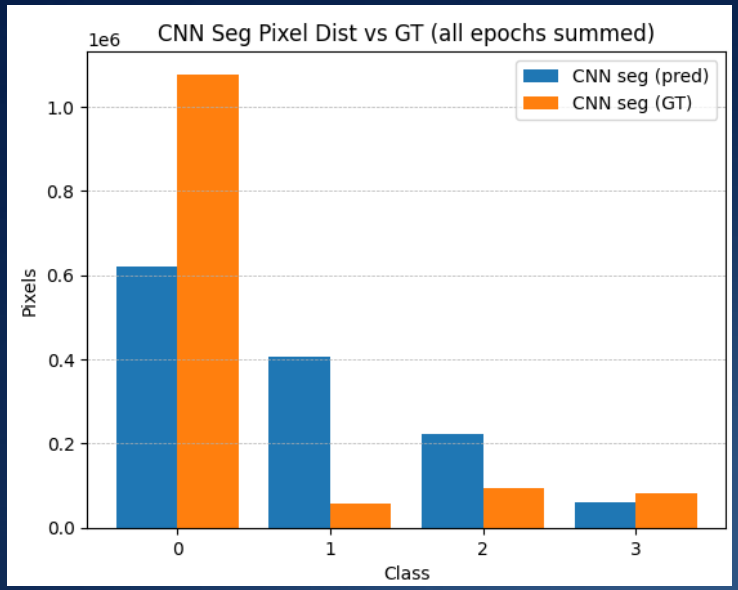


Comparison

BlendED

Hybrid QCNN model VS Classical CNN-Only model





{0: Background / 1: Small book / 2: Mid book / 3: Big book}

Comparison

BlendED

Hybrid QCNN model VS Classical CNN-Only model

<Same Evaluation>

Sample Batch of 30 Images
->Prediction of pixel and class

- Classification accuracy: 90.00%
- Pixel accuracy: 94.36%
- Mean IoU: 70.56%

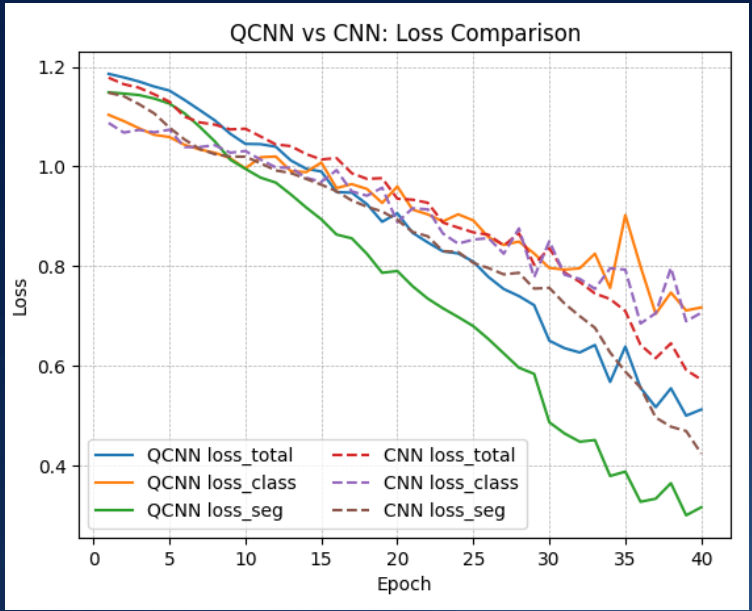
- Classification accuracy: 56.67%
- Pixel accuracy: 80.79%
- Mean IoU: 39.18%

Reasoning

BlendED

<Improvements>

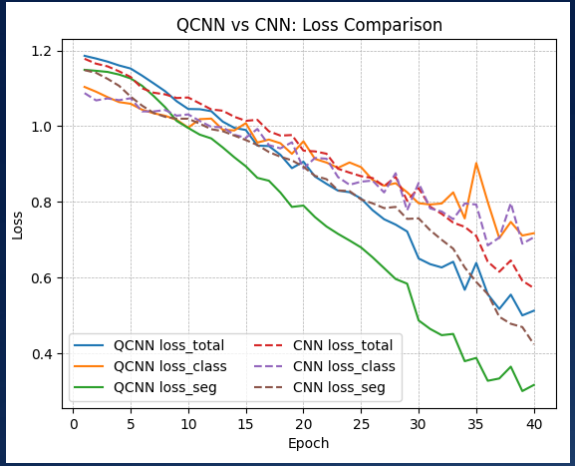
1. Quantum layer has introduced remarkable **improvement in segmentation**
2. Also, quantum layer has introduced remarkable **improvement in classification**
3. **Segmentation loss** has decreased drastically
4. **IoU**(Intersection over Union) has almost **doubled** in hybrid QCNN model



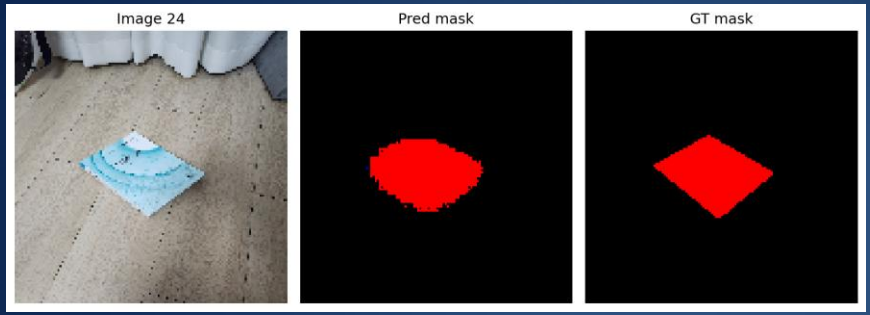
Reasoning

<Problems>

1. Trend of **classification loss** during the training are almost equivalent
2. Each **weights still need to be adjusted** to get higher accuracy, but also suppressing catching noises
3. Hybrid QCNN is much **slower** than the other, at this moment
4. **In some trials, classification was so erronous that it caught non-existing class in the object image, and did not match with segmentation**



sample_24: cls_head_pred=small_box(p=0.49), seg_dom_pred=big_box, seg_dom_true=big_box, cls_true=big_box
sample_25: cls_head_pred=small_box(p=0.42), seg_dom_pred=big_box, seg_dom_true=big_box, cls_true=big_box



Further Works

BlendED

1. Implementation on Real Hardware
    - Only accessible by cloud
    - Average waiting queue of minutes~hours (or even days...)
    - If iterative access at once is possible, this work would be applicable
2. Speed up
    - Assume QCNN Layer parameters depend on time, imaginary time evolution could substitute iterative waiting queue
    - Then, this might show meaningful speed ups

1. Point Cloud Extraction with Depth Estimation
    - As seen by YOLO, U-Net, the above mentioned jobs could be possible

2. PCA on the Object for Pose Estimation
    - Pipelining to PCA and object posing, this could be a complete process containing quantum computation

Discussion

Discussion

BlendED

1. YOLO, U-Net CNNs
    - Trying out existing classical models to experimentally observe image segmentation and classification
    - Comparison of classical models leading image segmenting CNN to be flexible from modification
    - Extension of classical CNN model to Quantum model

1. Hybrid QCNN - CNN model
    - Built up a model that has both CNN and QCNN conducting segmentation and classification
    - Quantum layer includes feature encoding, as well as parametrized circuit to introduce non-linear characteristics to classical model resulting in better performance on segmentation

2. Further Proposals
    - Iterative circuit measurement requires huge amount of waiting time on IBM platform queue, this problem could be further handled (or might be mitigated by technology develpment)
    - Since we have shown that applying quantum layer results in better accuracy, this method could be extended to complex segmentation jobs

References

1. Long, C., Huang, M., Ye, X., Futamura, Y., & Sakurai, T. (2025). Hybrid quantum-classical-quantum convolutional neural networks. Scientific Reports, 15, 31780. https://doi.org/10.1038/s41598-025-13417-1

1. Lu, Q., Lin, Y., & Yin, L. (2025). Quantum-enhanced deep lightweight object detection model for power tower base occlusion detection. Engineering Applications of Artificial Intelligence, 160, 111972. https://doi.org/10.1016/j.engappai.2025.111972

1. Russo, L., Mauro, F., Memar, B., Sebastianelli, A., Ullo, S. L., & Gamba, P. (2025). A quantum-assisted Attention U-Net for building segmentation over Tunis using Sentinel-1 data. arXiv:2507.13852. https://arxiv.org/abs/2507.13852

The End of Presentation